



COIN 496

Senior Project Portfolio Review

Friday, February 26th 2010

Dave Coons



Senior Portfolio

Dave Coons

COIN 332: Applied Networking

DJ Store (Final Project)

Grade Received: A

This simple ASP based website simulates a store selling DJ equipment. The site interacts with a Microsoft Access-type database for tracking customer information and invoices. To operate the 'store', the files must be loaded onto a web-server supporting ASP.



DAVE'S DJ STORE

Current Total: \$0.00

[View Cart](#)

Amplifier

\$399.00



[Add to Cart](#)

CD Turntables

\$699.00



[Add to Cart](#)

Crossover

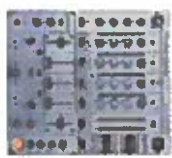
\$499.00



[Add to Cart](#)

Mixer

\$899.00



[Add to Cart](#)

Speakers

\$1,099.00



[Add to Cart](#)

```
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="Default.aspx.vb" Inherits="
    "_Default" %>
<%@ Import Namespace="System.Data" %>
<%@ Import Namespace="System.Data.OleDb" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/
    xhtml1/DTD/xhtml1-transitional.dtd">

<script runat="server">
    Dim Item As String
    Dim UserID As Integer
    Dim CustomerID As Integer
    Dim sConnString As String
    Dim oConn As OleDbConnection
    Dim eMail As String

    Sub Page_Load(ByVal Sender As Object, ByVal E As EventArgs)
        Item = Request("Item")
        UserID = Request("UserID")

        sConnString = "Provider=Microsoft.Jet.OLEDB.4.0; Data Source=" & Server.MapPath(
"database.mdb")
        oConn = New OleDbConnection(sConnString)

        oConn.Open()

    End Sub

    Sub Page_UnLoad(ByVal Sender As Object, ByVal E As EventArgs)
        oConn.Close()
    End Sub

    'Function GetDBValue(ByVal sSql As String, ByVal Field As String) As String
    '    Dim oCmd As OleDbCommand
    '    Dim oReader As OleDbDataReader

    '    oCmd = New OleDbCommand(sSql, oConn)
    '    oReader = oCmd.ExecuteReader()

    '    oReader.Read()

    '    If oReader.HasRows() And Not Field Is Nothing Then
    '        GetDBValue = oReader.Item(Field)
    '    Else
    '        GetDBValue = Nothing
    '    End If
    '    oCmd.Dispose()
    'End Function

    Sub Verify_User(ByVal sender As Object, ByVal e As System.EventArgs)
        Dim CustomerID As String

        eMail = txtEmailAddress.Text()

        CustomerID = GetDBValue("select CustomerID from tblCustomer where Email='" &
eMail & "';", "CustomerID", oConn)

        If (CustomerID Is Nothing) Then
            spanMessage.InnerHtml = "<p style='font-size:xx-large;'><strong>User email " &
& eMail & " not found</strong></p>"
        Else
            If GetDBValue("select UserPassword from tblCustomer where CustomerID=" &
CustomerID, "UserPassword", oConn) = txtPassword.Text() Then

                oConn.Close()
```

```
        Response.Redirect("Checkout.aspx?CustomerID=" & CustomerID & "&UserID=" &
        UserID.ToString())
    Else
        spanMessage.InnerHtml = "<p style='font-size:xx-large;'><strong>Password
is incorrect!</strong></p>"
    End If
End If

End Sub

</script>

<html xmlns="http://www.w3.org/1999/xhtml" >
<head runat="server">
    <title>Daves DJ Store</title>
</head>
<body onload="Page_Load" runat="server" onunload="Page_Unload">
    <form id="form1" runat="server">
        Please enter your email <asp:TextBox id="txtEmailAddress" runat="server"/><br />
        Please enter your password <asp:TextBox id="txtPassword" runat="server"/><br />
        <asp:Button onclick="Verify_User" runat="server" Text="Log In" /><br />
        or click <a href="register.aspx?UserID=<% Response.Write(UserID) %>">HERE</a> to
        register
    </form>
    <span id="spanMessage" runat="server"></span>
</body>
</html>
```

```

<%@ Page Language="VB" AutoEventWireup="false" CodeFile="Default.aspx.vb" Inherits="
    "_Default" %>
<%@ Import Namespace="System.Data" %>
<%@ Import Namespace="System.Data.OleDb" %>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/
    xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml" >
<head id="Head" runat="server">
    <title>Daves DJ Store</title>
    <script runat="server">
        Dim Item As String
        Dim UserID As Integer
        Dim CustomerID As Integer
        Dim sConnString As String
        Dim oConn As OleDbConnection
        Dim eMail As String

        Sub Page_UnLoad(ByVal Sender As Object, ByVal E As EventArgs)
            oConn.Close()
        End Sub

        Sub Page_Load(ByVal Sender As Object, ByVal E As EventArgs)

            sConnString = "Provider=Microsoft.Jet.OLEDB.4.0; Data Source=" & Server.
MapPath("database.mdb")
            oConn = New OleDbConnection(sConnString)

            oConn.Open()

            If Page.IsPostBack Then
                CreateCust()
            Else
                Item = Request("Item")
                UserID = Request("UserID")
            End If
        End Sub

        Sub CreateCust()
            Dim sSql As String
            Dim CustomerID As String

            UserID = Request("UserID")

            sSql = "INSERT into tblCustomer"
            sSql += " (Email,Name,Address,City,State,Zip,Phone,CCType,CCNumber,CCExpMonth
, CCExpYear) "
            sSql += " values ('" & Request.Form("txtEmail") & "','"
            sSql += "'" & Request.Form("txtCustFirstName") & " " & Request.Form(
"txtCustLastName") & "','"
            sSql += "'" & Request.Form("txtCustAddress") & "','"
            sSql += "'" & Request.Form("txtCustCity") & "','"
            sSql += "'" & Request.Form("txtCustState") & "','"
            sSql += "'" & Request.Form("txtCustZip") & "','"
            sSql += "'" & Request.Form("txtCustPhone") & "','"
            sSql += "'" & Request.Form("rbCreditCardType") & "','"
            sSql += "'" & Request.Form("txtCreditCardNumber") & "','"
            sSql += "'" & Request.Form("lbExpMonth") & "','"
            sSql += "'" & Request.Form("lbExpYear") & "');"

            GetDBValue(sSql, Nothing, oConn)

            sSql = "Update tblCustomer set UserPassword = '" & Request.Form("txtPassword"
) & "'" where Email = '" & Request.Form("txtEmail") & "'"

            GetDBValue(sSql, Nothing, oConn)

```



```

    }

    return(bAllFieldsValid);
}

//-->
</script>
</head>

<body onload="Page_Load" runat="server">
<h1>Please fill in the following fields</h1>
<form method="post" name="frmOrderForm" id="frmOrderForm" runat="server">
<p>First Name: &nbsp;<input name="txtCustFirstName" type="text" id="txtCustFirstName"
    value="First Name" /> &nbsp;<input name="txtCustLastName" type="text" id="txtCustLastName" value="Last Name" /></p>
<p>Street Address:&nbsp;<input name="txtCustAddress" type="text" id="txtCustAddress"
    size="50" value="Address" /></p>
<p>City:&nbsp;<input name="txtCustCity" type="text" id="txtCustCity" value="City" /> &nbsp;<input name="txtCustState" type="text" id="txtCustState" size="10"
    maxlength="2" value="State" /> &nbsp;<input name="txtCustZip" type="text" id="txtCustZip" size="10" maxlength="5" value="Zip" /></p>
<p>Phone:&nbsp;<input name="txtCustPhone" type="text" id="txtCustPhone" maxlength="10" value="Phone" /> &nbsp;</p>
<p>Payment Method? <input name="rbCreditCardType" type="radio" value="Visa" /> Visa
    <input name="rbCreditCardType" type="radio" value="MasterCard" /> MasterCard
    <input name="rbCreditCardType" type="radio" value="AmericanExpress" /> American Express</p>
<p>Credit Card Number &nbsp;<input name="txtCreditCardNumber" type="text" size="50"
    maxlength="16" /></p>
<p>Expiration Month: &nbsp;<select name="lbExpMonth" size="1">
    <option>&lt;Select Month&gt;</option>
    <option value="01">01</option>
    <option value="02">02</option>
    <option value="03">03</option>
    <option value="04">04</option>
    <option value="05">05</option>
    <option value="06">06</option>
    <option value="07">07</option>
    <option value="08">08</option>
    <option value="09">09</option>
    <option value="10">10</option>
    <option value="11">11</option>
    <option value="12">12</option></select>
    &nbsp;<select name="lbExpYear" id="lbExpYear">
    <option>&lt;Select Year&gt;</option>
    <option value="2007">2007</option>
    <option value="2008">2008</option>
    <option value="2009">2009</option>
    <option value="2010">2010</option>
    <option value="2011">2011</option></select></p>
<p>Email Address: &nbsp;<input type="text" id="txtEmail" name="txtEmail" /></p>
<p>Password: &nbsp;<input type="text" id="txtPassword" name="txtPassword" /></p>
<input name="pbSubmit" type="submit" value="Submit Info" onclick="return validateForm();" />
</form>
</body>
</html>

```



```

<%@ Page Language="VB" AutoEventWireup="false" CodeFile="Default.aspx.vb" Inherits=
    "_Default" %>
<%@ Import Namespace="System.Data" %>
<%@ Import Namespace="System.Data.OleDb" %>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/
    xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml" >
<head id="Head" runat="server">
    <title>Dave's DJ Store</title>
    <script runat="server" language="vb">
        Dim sConnString As String
        Dim oConn As System.Data.OleDb.OleDbConnection
        Dim UserID As Integer

        Sub Page_Load(ByVal Sender As Object, ByVal E As EventArgs)

            Dim InvoiceNum As String
            Dim InvoiceTotal As Double

            sConnString = "Provider=Microsoft.Jet.OLEDB.4.0; Data Source=" & Server.
MapPath("database.mdb")
            oConn = New OleDbConnection(sConnString)
            oConn.Open()
            UserID = Request("UserID")
            If UserID = 0 Then
                Randomize()
                UserID = Int(Rnd() * 65535)
                InvoiceTotal = "0.00"
            Else
                InvoiceNum = GetDBValue("SELECT [Invoice Number] as INV from tblInvoice
where UserID=" & UserID.ToString(), "INV", oConn)

                InvoiceTotal = Convert.ToDouble(GetRunningTotal(oConn, InvoiceNum))
            End If

            spnTotal.InnerHtml = "<p align='center'>Current Total: " & InvoiceTotal.
ToString("C2") & "</p>"

        End Sub
    </script>
</head>
<body onload="Page_Load" runat="server" >
    <p align="center">D A V E ' S &nbsp; &nbsp; D J &nbsp; &nbsp; S T O R E</p>
    <span id="spnTotal" runat="server"></span>
    <form id="frmMain" runat="server">
        <%
            Dim sSql As String
            Dim oCmd As OleDbCommand
            Dim oReader As OleDbDataReader
            Dim PartNumber As Integer
            Dim Price As Double
            Dim Desc As String
            Dim TableString As String

            Response.Write("<p align='center'><a href='purchase.aspx?UserID=" & UserID.
ToString() & "'><img src='ViewCart.gif'></a></p>")

            TableString = "<table cellpadding='20' style='border:groove; margin-left:auto
; margin-right:auto;'>"

            sSql = "Select * from tblPartNumber order by [Part Desc]"
            oCmd = New OleDbCommand(sSql, oConn)
            oReader = oCmd.ExecuteReader()

            Response.Write(TableString)

```

```
While oReader.Read()  
    PartNumber = oReader.Item("Part Number")  
    Desc = oReader.Item("Part Desc")  
    Price = oReader.Item("Price")  
    Response.Write("<tr>")  
    Response.Write("<td>" & Desc & "</td>")  
    Response.Write("<td>" & Price.ToString("C2") & "</td>")  
    Response.Write("<td align='center'><img src='" & PartNumber & ".jpg'></  
td>")  
    Response.Write("<td><a href='purchase.aspx?item=" & PartNumber & "&UserID=  
=" & UserID & "'><img src='addButton.jpg'></a></td>")  
    Response.Write("</tr>")  
  
End While  
Response.Write("</table>")  
  
oConn.Close()  
oConn = Nothing  
%>  
  
</form>  
<!-- <table cellpadding='5' style=' border:groove; auto; margin-right:auto;'><tr><td >  
</td></tr></table> -->  
</body>  
</html>
```

```

<!--% Page Language="VB" AutoEventWireup="false" CodeFile="Default.aspx.vb" Inherits=
    "_Default" %-->
<!--% Import Namespace="System.Data" %>
<!--% Import Namespace="System.Data.OleDb" %-->
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/
    xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml" >
<head id="Head" runat="server">
    <title>Dave's DJ Store</title>
    <script runat="server" language="vb">
        Dim sConnString As String
        Dim oConn As System.Data.OleDb.OleDbConnection
        Dim UserID As Integer

        Sub Page_Load(ByVal Sender As Object, ByVal E As EventArgs)

            Dim InvoiceNum As String
            Dim InvoiceTotal As Double

            sConnString = "Provider=Microsoft.Jet.OLEDB.4.0; Data Source=" & Server.
MapPath("database.mdb")
            oConn = New OleDbConnection(sConnString)
            oConn.Open()
            UserID = Request("UserID")
            If UserID = 0 Then
                Randomize()
                UserID = Int(Rnd() * 65535)
                InvoiceTotal = "0.00"
            Else
                InvoiceNum = GetDBValue("SELECT [Invoice Number] as INV from tblInvoice
where UserID=" & UserID.ToString(), "INV", oConn)

                InvoiceTotal = Convert.ToDouble(GetRunningTotal(oConn, InvoiceNum))
            End If

            spnTotal.InnerHtml = "<p align='center'>Current Total: " & InvoiceTotal.
ToString("C2") & "</p>"

        End Sub
    </script>
</head>
<body onload="Page_Load" runat="server" >
    <p align="center">D A V E ' S &nbsp; &nbsp; &nbsp; D J &nbsp; &nbsp; &nbsp; S T O R E</p>
    <span id="spnTotal" runat="server"></span>
    <form id="frmMain" runat="server">
        <%
            Dim sSql As String
            Dim oCmd As OleDbCommand
            Dim oReader As OleDbDataReader
            Dim PartNumber As Integer
            Dim Price As Double
            Dim Desc As String
            Dim TableString As String

            Response.Write("<p align='center'><a href='purchase.aspx?UserID=" & UserID.
ToString() & "'><img src='ViewCart.gif'></a></p>")

            TableString = "<table cellpadding='20' style='border:groove; margin-left:auto;
; margin-right:auto;'>"

            sSql = "Select * from tblPartNumber order by [Part Desc]"
            oCmd = New OleDbCommand(sSql, oConn)
            oReader = oCmd.ExecuteReader()

            Response.Write(TableString)

```

```
While oReader.Read()  
    PartNumber = oReader.Item("Part Number")  
    Desc = oReader.Item("Part Desc")  
    Price = oReader.Item("Price")  
    Response.Write("<tr>")  
    Response.Write("<td>" & Desc & "</td>")  
    Response.Write("<td>" & Price.ToString("C2") & "</td>")  
    Response.Write("<td align='center'><img src='" & PartNumber & ".jpg'></  
td>")  
    Response.Write("<td><a href='purchase.aspx?item=" & PartNumber & "&UserID=  
=" & UserID & "'><img src='addButton.jpg'></a></td>")  
  
    Response.Write("</tr>")  
  
End While  
Response.Write("</table>")  
  
oConn.Close()  
oConn = Nothing  
  
>  
  
</form>  
<!-- <table cellpadding='5' style=' border:groove; auto; margin-right:auto;'><tr><td >  
</td></tr></table> -->  
</body>  
</html>
```

```
<%@ Page Language="VB" %>
```

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/
xhtml1/DTD/xhtml1-transitional.dtd">
```

```
<script runat="server">
```

```
</script>
```

```
<html xmlns="http://www.w3.org/1999/xhtml" >
```

```
<head runat="server">
```

```
<title>Untitled Page</title>
```

```
</head>
```

```
<body>
```

```
<form id="form1" runat="server">
```

```
<h1>
```

```
<%
```

```
    If Request("Status") = "Aborted" Then
        Response.Write("PURCHASE CANCELED")
```

```
    Else
```

```
        Response.Write("Purchase Confirmed")
```

```
    End If
```

```
%>
```

```
</h1>
```

```
</form>
```

```
</body>
```

```
</html>
```

```

<%@ Page Language="VB" AutoEventWireup="false" CodeFile="Default.aspx.vb" Inherits=
    "_Default" %>
<%@ Import Namespace="System.Data" %>
<%@ Import Namespace="System.Data.OleDb" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/
    xhtml1/DTD/xhtml1-transitional.dtd">

<script runat="server">
    Dim Item As String
    Dim UserID As Integer
    Dim CustomerID As Integer
    Dim sConnString As String
    Dim oConn As OleDbConnection
    Dim eMail As String
    Dim InvoiceNum As Integer

    Sub Purchase(ByVal Sender As Object, ByVal e As System.Web.UI.WebControls.
        CommandEventArgs)
        Dim sSql As String
        Dim InvoiceTotal As Double

        InvoiceTotal = GetDBValue("SELECT Sum(tblPartNumber.Price) AS InvTotal FROM
tblInvoice INNER JOIN (tblPartNumber INNER JOIN tblInvoiceLine ON tblPartNumber.[Part
    Number] = tblInvoiceLine.PartNumber) ON tblInvoice.[Invoice Number] = tblInvoiceLine
.[Invoice Number] GROUP BY tblInvoice.[Invoice Number] HAVING tblInvoice.[Invoice
    Number]=" & InvoiceNum.ToString(), "InvTotal", oConn)

        GetDBValue("UPDATE tblInvoice SET CustomerID = ' ' & CustomerID & ' ' where
[Invoice Number] = " & InvoiceNum & ";", Nothing, oConn)

        GetDBValue("UPDATE tblInvoice SET UserID = NULL where [Invoice Number] = " &
InvoiceNum & ";", Nothing, oConn)
        GetDBValue("UPDATE tblInvoice SET Shipping = " & (InvoiceTotal * 0.05) & " where
[Invoice Number] = " & InvoiceNum & ";", Nothing, oConn)
        GetDBValue("UPDATE tblInvoice SET ItemTotal = " & (InvoiceTotal * 1.05) & " where
[Invoice Number] = " & InvoiceNum & ";", Nothing, oConn)

        Response.Redirect("Complete.aspx?Status=Success&UserID=" & Request("UserID"))
    End Sub

    Sub Cancel(ByVal Sender As Object, ByVal e As System.Web.UI.WebControls.
        CommandEventArgs)

        UserID = Request("UserID")

        GetDBValue("UPDATE tblInvoice set UserID = ' ' where UserID" & UserID, Nothing,
oConn)

        Response.Redirect("Complete.aspx?Status=Aborted&UserID=" & Request("UserID"))
    End Sub

    Sub Page_UnLoad(ByVal Sender As Object, ByVal E As EventArgs)
        oConn.Close()
    End Sub

    Sub Page_Load(ByVal Sender As Object, ByVal E As EventArgs)
        Dim sSql As String

        sConnString = "Provider=Microsoft.Jet.OLEDB.4.0; Data Source=" & Server.MapPath(
"database.mdb")
        oConn = New OleDbConnection(sConnString)

```

```

oConn.Open()

UserID = Request("UserID")
CustomerID = Request("CustomerID")

sSql = "Select [Invoice Number] as InvNum from tblInvoice where UserID = " &
UserID
InvoiceNum = GetDBValue(sSql, "InvNum", oConn)

End Sub

</script>

<html xmlns="http://www.w3.org/1999/xhtml" >
<head runat="server">
<title>Dave's DJ Store</title>
</head>
<body onload="Page_Load" onunload="Page_unLoad" runat="server">
<form id="form1" runat="server">
<%
Dim sSql As String
Dim oCmd As OleDbCommand
Dim oReader As OleDbDataReader
Dim InvoiceTotal As Integer
Dim UnitCost As Integer
Dim LineTotal As Integer
Dim CustomerName As String

CustomerName = GetDBValue("Select Name from tblCustomer where CustomerID = " &
CustomerID, "Name", oConn)

sSql = "SELECT [Part Number],[Part Desc], Count([Part Number]) AS Quantity, Sum
(Price)/Count([Part Number]) AS Each, Sum(Price) AS LineTotal FROM tblInvoice INNER
JOIN (tblPartNumber INNER JOIN tblInvoiceLine ON tblPartNumber.[Part Number] =
tblInvoiceLine.PartNumber) ON tblInvoice.[Invoice Number] = tblInvoiceLine.[Invoice
Number] GROUP BY tblInvoice.[UserID],[Part Number],[Part Desc],tblInvoice.[Invoice
Number],tblInvoiceLine.[Invoice Number] HAVING tblInvoice.[UserID]=' & UserID.
ToString()
oCmd = New OleDbCommand(sSql, oConn)
oReader = oCmd.ExecuteReader()

Response.Write("<p style='font-size:30px;' align='center'>Welcome " & CustomerName &
"!</p>")
Response.Write("<table cellpadding='20' style='margin-left:auto; margin-right:auto;'>")
Response.Write("<tr><td align='center'>Description</td><td align='center'>Quantity</td>
<td align='center'>Unit Cost</td><td align='center'>Line Total</td><td align='center'>
>Image</td>")
While oReader.Read()
UnitCost = oReader.Item("Each")
LineTotal = oReader.Item("LineTotal")
Response.Write("<tr>")
Response.Write("<td align='center'>" & oReader.Item("Part Desc") & "</td>")
Response.Write("<td align='center'>" & oReader.Item("Quantity") & "</td>")
Response.Write("<td align='center'>" & UnitCost.ToString("C2") & "</td>")
Response.Write("<td align='center'>" & LineTotal.ToString("C2") & "</td>")
Response.Write("<td align='center'><img src='" & oReader.Item("Part Number") & ".
jpg'></td>")
Response.Write("</tr>")
End While

Response.Write("<table cellpadding='10' style='margin-left:auto; margin-right:auto;'>")
InvoiceTotal = GetDBValue("SELECT Sum(tblPartNumber.Price) AS InvTotal FROM
tblInvoice INNER JOIN (tblPartNumber INNER JOIN tblInvoiceLine ON tblPartNumber.[Part
Number] = tblInvoiceLine.PartNumber) ON tblInvoice.[Invoice Number] = tblInvoiceLine

```

```
.[Invoice Number] GROUP BY tblInvoice.[Invoice Number] HAVING tblInvoice.[Invoice Number]=" & InvoiceNum.ToString(), "InvTotal", oConn) ✓
```

```
Response.Write("<tr><td>Sub Total</td><td align='right'>" & InvoiceTotal.ToString("C2") & "</td></tr>") ✓
```

```
Response.Write("<tr><td>Shipping (5%)</td><td align='right'>" & (InvoiceTotal * 0.05).ToString("C2") & "</td></tr>") ✓
```

```
Response.Write("<tr><td><strong>Total</strong></td><td align='right'><strong>" & (InvoiceTotal * 1.05).ToString("C2") & "<strong></td></tr>") ✓
```

```
Response.Write("</table>")
```

```
oCmd.Dispose()
```

```
>
```

```
<table cellpadding='20' style='margin-left:auto; margin-right:auto;' runat="server">
<tr>
<td colspan='2' align='center'><asp:Button ID="Button1" OnCommand="Purchase" Text="Confirm" runat="server"/></td> ✓
<td>&nbsp;</td>
<td colspan='2' align='center'><asp:Button OnCommand="Cancel" Text="Cancel" runat="server"/></td> ✓
</tr>
</table>
```

```
</form>
```

```
</body>
```

```
</html>
```


Senior Portfolio

Dave Coons

COIN 315: Data Structures

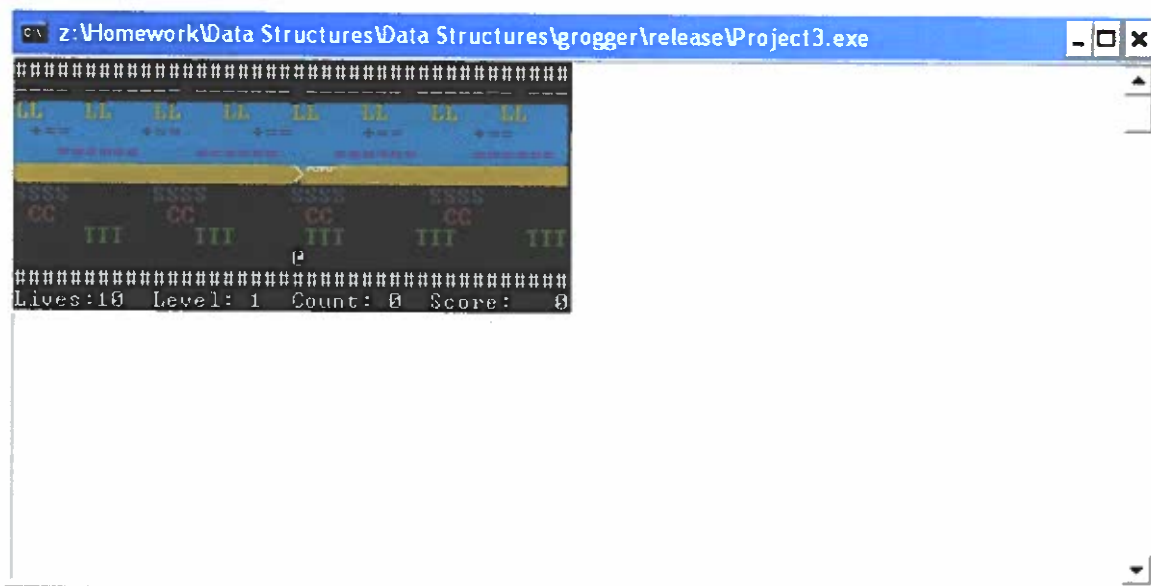
Grogger (Garry Underwood's Frogger port)

Grade Recieved: A

This simple 'game' was a console-based port of the popular 1980's game Frogger. It was designed as exercise in creating a stand-alone application. The program is written primarily in C, with a few C++ attributes such as classed and messaging. True C++ flavor, however, is missing. The program has been ported since originally written into Visual Studio 2008, and can be recompiled using the project file `grogger.sln`.

The program executable is actually called `Project3.exe`, since it was the third project submitted in this class. It is a simple DOS-style console application. Play is affected by moving the cursor keys in attempt to move the 'player' across the three lanes of traffic, onto the beach (avoiding the 'alligators mouth'), and onto the floating logs to finally lodge in one of the five open spaces at the top row.

As explained in the header for the file `world.c`, the program was written with double-buffering in mind. However, since this was not possible (realistic), there is an annoying 'flickering' during the animation. This is because each object is coded as a separate object, and is redrawn independently. A better approach would have been to treat each row as an object, perhaps with child-objects, allowing each row to be refreshed at once.



```

/*****
*****
*   Student Name:           Dave Coons
*
*   Program Name:           Grogger
*
*   Creation Date:          April 2009
*
*   Last Modified Date:     March 2009
*
*   COIN Course:            COIN 315: Data Structures
*
*   Grade Received:         A
*
*   Comments regarding design: In retrospect, each object being an 'object' created the
    annoying strobing *
*
*   double-buffered *
*
*   game. Each *
*
*   row should have been an object.
*
*****/
#include "CConsole.h"
#include "Water.h"
#include "World.h"
#include "Vehicle.h"
#include "Object.h"
#include "Landing.h"
#include <string>

#define FRAMES_PER_SECOND 30

static void CALLBACK ProcessTimer( UINT timerid, UINT msg, DWORD_PTR user,
    DWORD_PTR dw1, DWORD_PTR dw2 );
#define CALC_SPACE(n,i) ((int) (i*(SCREEN_WIDTH/(n))))

World::World(int theDelay)
{
    int idx = 0;

    int totalObjects = 0;

    bFirstPainting = true;

    TheScreen = new CConsole(SCREEN_WIDTH, SCREEN_HEIGHT+1);

    LivesLeft = 1;

    TicsPerRefresh = theDelay;
    Surfaces[idx++] = black;
    Surfaces[idx++] = black;
    Surfaces[idx++] = cyan;
    Surfaces[idx++] = cyan;
    Surfaces[idx++] = cyan;
    Surfaces[idx++] = brown;
    Surfaces[idx++] = black;
    Surfaces[idx++] = black;
    Surfaces[idx++] = black;
    Surfaces[idx++] = black;
    Surfaces[idx++] = black;

    // Set hitched to 0, since we are not riding anybody
    Hitched = NULL;

```

```
AllTheObjects = new Danger *[NUM_TRUCKS + NUM_CARS + NUM_SEMIS + NUM_BIGLOGS +
NUM_SHORTLOGS + NUM_GATORS + NUM_SNAKES];
```

```
for(idx = 0; idx< NUM_TRUCKS; idx++)
{
    AllTheObjects[totalObjects++] = TheTrucks[idx] =
        new Truck( 8,          // Row for these dangers
                  green,       // Foreground color
                  Surfaces[8], // Background color
                  13*theDelay, // Velocity
                  CALC_SPACE(NUM_TRUCKS, idx)+5);
}

for(idx = 0; idx< NUM_CARS; idx++)
{
    AllTheObjects[totalObjects++] = TheCars[idx] =
        new Car(7,          // Row for these dangers
               red,         // Foreground color
               Surfaces[7], // Background color
               -5*theDelay, // Velocity
               CALC_SPACE(NUM_CARS, idx)+2);
    //AllTheObjects[totalObjects++] = TheCars[idx];
}

for(idx = 0; idx< NUM_SEMIS; idx++)
{
    AllTheObjects[totalObjects++] = TheSemis[idx] =
        new Semi( 6,          // Row for these dangers
                blue,         // Foreground color
                Surfaces[6], // Background color
                7*theDelay,   // Velocity
                CALC_SPACE(NUM_SEMIS, idx));
}

for(idx = 0; idx< NUM_BIGLOGS; idx++)
{
    AllTheObjects[totalObjects++] = TheBigLogs[idx] =
        new FallingLog( 4,          // Row for these dangers
                      purple,       // Foreground color
                      Surfaces[4],   // Background color
                      7*theDelay,    // Velocity
                      CALC_SPACE(NUM_BIGLOGS, idx)+3);
}

for(idx = 0; idx< NUM_SHORTLOGS; idx++)
{
    AllTheObjects[totalObjects++] = TheShortLogs[idx] =
        new ShortLog( 2,          // Row for these dangers
                    brown,        // Foreground color
                    Surfaces[2],   // Background color
                    7*theDelay,    // Velocity
                    CALC_SPACE(NUM_SHORTLOGS, idx));
}

for(idx = 0; idx< NUM_GATORS; idx++)
{
    AllTheObjects[totalObjects++] = TheGators[idx] =
        new Gator( 3,          // Row for these dangers
                 blue,         // Foreground color
                 Surfaces[3],   // Background color
                 -8*theDelay,   // Velocity
                 CALC_SPACE(NUM_GATORS, idx)+1);
}

AllTheObjects[totalObjects++] = TheSnake =
```

```

        new Snake( 5,           // Row for this dangers
                   white,       // Foreground color
                   Surfaces[5], // Background color
                   -7*theDelay, // Velocity
                   SCREEN_WIDTH/2);

    Frogger = new Player(white, black, 0, 0);

    TopRow = new Landing(NUM_LANDINGS, 1);

    Level = 1;
}

World::~World()
{
    if (TheScreen) delete TheScreen;
}

void World::run()
{
    MMRESULT timerID;

    if (Frogger)
    {
        TIMECAPS tc;
        timeGetDevCaps(&tc, sizeof(TIMECAPS));
        DWORD resolution = min(max(tc.wPeriodMin, 0), tc.wPeriodMax);
        timeBeginPeriod(resolution);

        timerID = timeSetEvent( (1000/FRAMES_PER_SECOND),
                                resolution,
                                (LPTIMECALLBACK)ProcessTimer,
                                (DWORD_PTR)this,
                                TIME_PERIODIC);

        while (Frogger->GetLivesLeft())    TheScreen->delay(100);
    }

    timeKillEvent(timerID);

    // Wait for a few iterations to make sure no more timer calls are in the system
    TheScreen->delay(1000/FRAMES_PER_SECOND * 2);
}

void CALLBACK World::ProcessTimer( UINT timerid, UINT msg, DWORD_PTR user, DWORD_PTR dw1,
    DWORD_PTR dw2)
{
    char    keyInput = 0;
    World   *world;
    bool    bDead = false;

    world = (World *)user;
    if (world)
    {
        if (world->Frogger->GetLivesLeft())
        {
            world->MoveObjects();
            world->Frogger->Move(world->TheScreen, world->TopRow, world->Hitched);

            world->Frogger->UpdateBackgroundColor(world->Surfaces[world->Frogger->GetRow
    ());
}

```

```

        world->DetectRiding();

        if (world->DetectDeath() || bDead)
            world->Frogger->KillOneLife();

        world->Paint();

        // Should we move up one level?
        if (world->TopRow->NumberOfLandingsUsed() >= NUM_LANDINGS)
        {
            world->Level++;
            world->Frogger->ResetPosition(true);

            world->TopRow->Reset();
        }
    }
}

bool World::DetectDeath()
{
    bool    ret_val = false;

    // Check for being run over
    ret_val |= DetectCollision();

    // Check to see if we are in the water
    if ((Surfaces[Frogger->GetRow()] == cyan) && !Hitched)
        ret_val = true;

    // Did we get eaten?
    if (Hitched && Hitched->DangerEatUs())
        ret_val = true;

    return(ret_val);
}

void World::Paint()
{
    int idx = 0;

    TheScreen->setColor(white, Surfaces[idx++]); DrawLine(0, idx-1, SCREEN_WIDTH, '#') ✓
    ;
    TheScreen->setColor(white, Surfaces[idx++]); TopRow->Paint(TheScreen);
    {
        bool    rowsOfScreen[SCREEN_HEIGHT];

        // Clear out the array
        memset(rowsOfScreen, false, sizeof(rowsOfScreen));

        // Go through the objects, marking the rows that need to be updated
        for (idx = 0; idx < TOTAL_DANGERS; idx++)
        {
            if (AllTheObjects[idx]->DangerNeedsPainting() && AllTheObjects[idx]->
GetObjectType() != tPlayer) ✓
                rowsOfScreen[AllTheObjects[idx]->GetRow()] = true;

        }

        for(idx=2; idx < SCREEN_HEIGHT; idx++)
            if (rowsOfScreen[idx])
            {
                TheScreen->setColor(white, Surfaces[idx]);
                DrawLine(0, idx, SCREEN_WIDTH, ' ');
            }
        TheScreen->setColor(white, Surfaces[idx-1]); DrawLine(0, idx-1, SCREEN_WIDTH, ✓

```

```

    '#');
}

if (Hitched)
    Hitched->SetNeedsRepainting();

DrawObjects();

TheScreen->setColor(white, Surfaces[Frogger->GetRow()]);

Frogger->Paint(TheScreen);

// Update score and lives
{
    char score[150];
    TheScreen->setColor(white, black);
    sprintf_s(
        score,
        sizeof(score),
        "Lives:%2d Level:%2d Count:%2d Score:%4d",
        Frogger->GetLivesLeft(),
        Level,
        TopRow->NumberOfLandingsUsed(),
        Frogger->GetCurrentScore());
    TheScreen->printString(0, SCREEN_HEIGHT, score);
}

}

void World::DrawLine(int x, int y, int colsToDraw, char charToDraw)
{
    for (int idx=0; idx < colsToDraw; idx++)
        TheScreen->printChar(x+idx, y, charToDraw);
}

void World::DrawObjects()
{
    for (int idx = 0; idx < TOTAL_DANGERS; idx++)
        if (AllTheObjects[idx]->DangerNeedsPainting() && AllTheObjects[idx]->
            GetObjectType() != tPlayer)
            AllTheObjects[idx]->Paint(TheScreen);
}

void World::MoveObjects()
{
    // Move Vehicles
    for (int idx = 0; idx < TOTAL_DANGERS; idx++)
        AllTheObjects[idx]->Move();
}

bool World::DetectCollision()
{
    bool collision = false;

    for (int idx = 0; (idx < TOTAL_DANGERS) && !collision; idx++)
    {
        if (AllTheObjects[idx]->GetObjectType() & VEHICLE_DANGERS)
        {
            collision |= AllTheObjects[idx]->DetectTouching(Frogger->GetRow(), Frogger->
                GetColumn());
        }
    }
    return(collision);
}

bool World::DetectRiding()

```

```
{
    bool    ret_val = false;
    Danger  *newHitched = NULL;

    for (int idx = 0; (idx < TOTAL_DANGERS) && !newHitched; idx++)
    {
        if (AllTheObjects[idx]->GetObjectType() & RIDING_DANGERS)
        {
            if (AllTheObjects[idx]->DetectTouching(Frogger->GetRow(), Frogger->GetColumn()
            ()))
                newHitched = AllTheObjects[idx];
        }
    }
    if (newHitched)
    {
        Hitched = newHitched;

        int frogPosition = 0;

        frogPosition = Frogger->GetColumn() - Hitched->GetColumn();

        // If we wrapped, adjust for it
        if (frogPosition < 0)
            frogPosition += SCREEN_WIDTH;

        Hitched->SetFrogPosition(frogPosition);
    } else
    {
        // Redraw the danger since we hopped off
        if (Hitched)    Hitched->SetNeedsRepainting();

        Hitched = NULL;
    }

    return(ret_val);
}
```



```

#ifndef _WORLD
#define _WORLD
#include "Globals.h"
#include "Player.h"
#include "Vehicle.h"
#include "Object.h"
#include "Water.h"
#include "Landing.h"

class World
{
public:
    World(int theDelay);
    ~World();
    void run(); // Starts the game
private:
    int Level; // Current Level
    Landing *TopRow; // The destination of Frogger;
    void Die(); // Ends the game
    Danger *Hitched; // Pointer to the Danger the frog is riding, or NULL
    int GetRow(); // returns row of current Danger
    int Cols; // Cols on the screen
    int Rows; // Number of rows of the screen
    Color Surfaces[SCREEN_HEIGHT]; // Array of surfaces representing each row
    int TicsPerRefresh; // Number of system tics per update
    void Paint(); // Redraw the screen
    CConsole *TheScreen; // The output screen
    Player *Frogger; // The player
    void DrawLine(int x, int y, int colsToDraw, char charToDraw); // Draw a horizontal line with the specified char to theScreen
    void DrawObjects(); // Draw the dangers
    void MoveObjects(); // Move the dangers
    bool DetectCollision(); // Check for being run over
    bool DetectRiding(); // Check to see if we hitched a ride
    bool DetectDeath(); // Check to see if the player died
    int RowNeedsPainting(Danger *o); // Returns row of Danger if it needs repainting, or -1
    int LivesLeft; // Status of the player
    Truck *TheTrucks[NUM_TRUCKS]; // Array of trucks
    Car *TheCars[NUM_CARS]; // Array of cars
    Semi *TheSemis[NUM_SEMIS]; // Array of Semis
    FallingLog *TheBigLogs[NUM_BIGLOGS]; // Array of falling logs
    ShortLog *TheShortLogs[NUM_SHORTLOGS]; // Array of short logs
    Gator *TheGators[NUM_GATORS]; // Array of short logs
    Snake *TheSnake;
    Danger **AllTheObjects; // Dynamic array of all the objects
    static void CALLBACK World::ProcessTimer( UINT , UINT , DWORD_PTR , DWORD_PTR , DWORD_PTR);
private:
    bool bFirstPainting; // True if we haven't painted background
};
#endif

```

```
#include "Globals.h"
#include "CConsole.h"
#include "Player.h"
```

```
Player::Player(Color fColor, Color bColor, int dir, int velocity):Object(0, 0, (int)
    strlen(FROG_PRESENTATION), fColor, bColor)
```

```
{
    Type = tPlayer;
    strcpy_s(Representation, sizeof(Representation), FROG_PRESENTATION);
    currentLives = PLAYER_LIVES;
    ResetPosition(false);
    Score = 0;
}
```

```
void Player::Move(CConsole *theScreen, Landing *topRow, Danger *carrier)
```

```
{
    char    keyInput;
    int moveX = 0, moveY = 0;

    if (theScreen->getInput(keyInput))
    {
        if (keyInput == '8')    moveY = -1;
        else if (keyInput == '2')    moveY = 1;
        else if (keyInput == '4')    moveX = -1;
        else if (keyInput == '6')    moveX = 1;

        // Paint over the last position
        theScreen->setColor(FgColor, BgColor);
        theScreen->printChar(Column, Row, ' ');
    }
    if (carrier)
    {
        // This means the Danger moved one space
        if ((carrier->GetColumn() + carrier->GetFrogPosition()) != Column)
        {
            if ((carrier->GetObjectType() != tBigLog) || (Column < (SCREEN_WIDTH-1)))
            {
                int newColumn;

                // Move the frog
                newColumn = carrier->GetColumn() + carrier->GetFrogPosition();

                if (newColumn >= SCREEN_WIDTH)    newColumn = 0;
                else if (newColumn < 0)    newColumn = SCREEN_WIDTH-1;

                Column = newColumn;
            }
            else if ((carrier->GetObjectType() == tBigLog) && (Column >= (SCREEN_WIDTH-
1)))
                KillOneLife();
        }
    }

    if (moveY)
    {
        // bNeedsRepainting = true;

        if ((Row+moveY) < HighestRowReached)
        {
            HighestRowReached--;
            Score++;
        }

        Row += moveY;
        if (Row < 1)
    }
}
```

```
        Row = 1;
        if (Row >= (SCREEN_HEIGHT - 1))
            Row = SCREEN_HEIGHT - 2;
    }

    if (moveX)
    {
        Object::Move(moveX);
    }

    if (Row == 1)
    {
        if (!topRow->LandFrog(Column))
        {
            //bDead = true;
            KillOneLife();
        }
        else
            ResetPosition(true);
    }
}

void Player::ResetPosition(bool bUpdateLevel)
{
    Row = SCREEN_HEIGHT-2;
    Column = SCREEN_WIDTH/2;
    HighestRowReached = Row;

    if (bUpdateLevel)    Score+=10;
}

void Player::Paint(CConsole *screen)
{
    screen->printString((int)Column, Row, FROG_PRESENTATION);
    // bNeedsRepainting = false;
}

int Player::GetColumn()
{
    return((int)Column);
}

int Player::GetRow()
{
    return(Row);
}

void Player::KillOneLife()
{
    currentLives--;
    ResetPosition(false);
}

int Player::GetLivesLeft()
{
    return (currentLives);
}
```

```
#ifndef _PLAYER
#define _PLAYER
#include "Danger.h"
#include "Landing.h"

#define FROG_PRESENTATION "@"
#define PLAYER_LIVES 10

class Player : public Object
{
private:
    int currentLives;
    int Score;
    int HighestRowReached;
public:
    Player(Color fColor, Color bColor, int dir, int velocity);
    void ResetPosition(bool bUpdateLevel);
    void Move(CConsole *theScreen, Landing *topRow, Danger *carrier);
    void Paint(CConsole *screen);
    int GetRow(); // returns row of current object
    int GetColumn(); // returns column of current object
    void KillOneLife();
    int GetLivesLeft();
    int GetCurrentScore() {return (Score*5);};
    bool DangerEatUs() {return false;};
};
#endif
```

```
#include "Globals.h"
#include "Object.h"
#include "Danger.h"

void Object::Move(int x)
{
    Column += x;

    if (Column > (SCREEN_WIDTH-1)) Column = 0;
    if (Column < 0) Column = (SCREEN_WIDTH-1);

    bNeedsRepainting = true;
}

void Object::Paint(CConsole *theScreen)
{
    int pRow, pCol;

    pRow = (int)(Row );
    pCol = (int)(Column + 0);

    theScreen->setColor(FgColor, BgColor);
    if (pCol >= 0 && pCol <= (SCREEN_WIDTH - Width))
        theScreen->printString(pCol, Row, Representation);
    // We need to wrap to the front of the screen
    else if (pCol > (SCREEN_WIDTH - Width))
    {
        char buf[50];

        // Copy text to local buffer
        strcpy_s(buf, sizeof(buf), Representation);

        // terminate excess string
        buf[-(int)Column] = '\0';

        buf[SCREEN_WIDTH-pCol] = '\0';

        // Print the trailing edge on the screen, clipping the vehicle presentation
        theScreen->printString(pCol, pRow, buf);

        // Copy text to local buffer
        strcpy_s(buf, sizeof(buf), (Representation+(SCREEN_WIDTH-pCol)));

        // Print the other leading edge on the other side
        theScreen->printString(0, pRow, buf);
    }
    bNeedsRepainting = false;
}
```

```

#ifndef _OBJECT
#define _OBJECT
#include "Globals.h"
#include "CConsole.h"

#define SIGN(x) ((x == 0) ? 0 : (x)/abs(x))

enum ObjectType
{
    tPlayer    = 1,
    tTruck     = 2,
    tCar        = 4,
    tSemi       = 8,
    tBigLog     = 16,
    tShortLog   = 32,
    tGator      = 64,
    tSnake      = 128
};

class Object
{
protected:
    ObjectType    Type;           // Type of object
    int           Row;            // The row the Danger lives on
    int           Column;         // The column the Danger lives on
    Color         FgColor;        // The foreground color of the Danger
    Color         BgColor;        // The background color of the Danger
    bool          bNeedsRepainting; // True if Danger has moved
    int           Width;          // The Dangers width
    char          Representation[50]; // String to represent vehicle

public:
    Object();
    Object(int objRow,
           int objColumn,
           int objWidth,
           Color fColor,
           Color bColor)
    {
        Row = objRow; Column = objColumn;
        Width = objWidth;
        FgColor = fColor; BgColor = bColor;
        bNeedsRepainting = true;
    }

    virtual void UpdateBackgroundColor(Color bColor) {BgColor=bColor;};
    virtual int  GetRow() {return Row;} // returns row of current Danger
    virtual int  GetColumn() {return Column;} // returns column of current
    Danger
    virtual void Move(int x); // Moves the Danger x units along
    default direction. Also animates the character
    virtual void Paint(CConsole *screen); // Draw the Danger
    virtual void SetNeedsRepainting() {bNeedsRepainting = true;};
    virtual void GetObjectType() {return Type;};
};

#endif

```



```
}

void CConsole::printStringClearLine(int x, int y, const std::string& s) { //clears rest of
    line after string printed
    gotoXY(x,y);
    printStringClearLine(s);
}

void CConsole::printStringClearLine(const std::string& s) { //clears rest of line after
    string printed
    printString(s);
    CONSOLE_SCREEN_BUFFER_INFO info;
    GetConsoleScreenBufferInfo(m_hConsole,&info);
    for (int col=info.dwCursorPosition.X; col < m_width; col++)
        printChar(' '); //print spaces for rest of line, console b/c don't know if we
        started at 0
}

void CConsole::printInt(int x, int y, int val) { //print an integer value
    gotoXY(x,y);
    printInt(val);
}

void CConsole::printInt(int val) { //print an integer value
    char buf[12]={0};
    _itoa_s(val, buf, 11, 10); //safely convert int to ascii text
    printString(buf); //constructor for string automatically called for char *
}
```



```
#pragma once
#include <string>
#include <conio.h>
#include <windows.h>
```

```
enum Color {black=0, blue=FOREGROUND_BLUE, green=FOREGROUND_GREEN, red=FOREGROUND_RED,
purple=FOREGROUND_BLUE + FOREGROUND_RED, cyan=FOREGROUND_BLUE+FOREGROUND_GREEN,
brown=FOREGROUND_GREEN+FOREGROUND_RED, white=FOREGROUND_BLUE+FOREGROUND_GREEN+
    FOREGROUND_RED};
```

```
class CConsole {
public:
    CConsole(int width, int height);
    void setColor(Color fore, Color back) {SetConsoleTextAttribute(m_hConsole, fore +
        (back<<4) );}
    bool getInput(char& ch);
    void waitEnter(); // wait until the user presses ENTER
    void gotoXY(int x, int y); //moves cursor to position x (from left), y (from top)
        //0,0 is upper left corner
    void printChar(int x, int y, char ch); //prints ch at x,y, moves cursor over 1 to
        right
    void printChar(char ch) {_putch(ch);} //prints ch where current cursor is, moves
        cursor over 1 to right
    void printString(int x, int y, const std::string& s);
    void printString(const std::string& s);
    void printStringClearLine(const std::string& s); //clears rest of line after string
        printed
    void printStringClearLine(int x, int y, const std::string& s);
    void printInt(int x, int y, int val); //print an integer value at x,y
    void printInt(int val); //print an integer value
    void delay(int amount) {Sleep(amount);}
private:
    HANDLE m_hConsole;
    int m_width;
    int m_height;
};
```

```
#include "CConsole.h"
#include "Landing.h"
#include "Globals.h"

Landing::Landing(int count, int destRow)
{
    NumZones = count;

    Row = destRow;

    LandingZones = new char[SCREEN_WIDTH];

    Reset();
}

Landing::~Landing()
{
    delete []LandingZones;
}

void Landing::Reset()
{
    // Clear out the landing zone array
    memset(LandingZones, TOP_ROW_WALL_CHAR, SCREEN_WIDTH);

    // Set up the requested landing zones
    for (int x=0; x < NumZones; x++)
        LandingZones[(int)((SCREEN_WIDTH / NumZones) * (x+.5))] = ' ';
}

void Landing::Paint(CConsole *theScreen)
{
    for (int x=0; x < SCREEN_WIDTH; x++)
    {
        theScreen->printChar(x, Row, LandingZones[x]);
    }
}

bool Landing::LandFrog(int col)
{
    bool    bFrogLanded;

    if (LandingZones[col] == TOP_ROW_WALL_FREE)
    {
        LandingZones[col] = TOP_ROW_WALL_OCCUPIED;
        bFrogLanded = true;
    }
    else bFrogLanded = false;

    return(bFrogLanded);
}

int Landing::NumberOfLandingsUsed(void)
{
    int LandingsFree = 0;

    for (int x=0; x<SCREEN_WIDTH; x++)
    {
        if (LandingZones[x] == TOP_ROW_WALL_FREE)    LandingsFree++;
    }

    return(NumZones - LandingsFree);
}
```

```
#ifndef _LANDING
#define _LANDING

#define TOP_ROW_WALL_CHAR    '-'
#define TOP_ROW_WALL_OCCUPIED 'X'
#define TOP_ROW_WALL_FREE    ' '

class Landing
{
public:
    Landing(int count, int destRow);
    ~Landing();
    bool    LandFrog(int col);
    void    Paint(CConsole *theScreen);
    int     NumberOfLandingsUsed();
    void    Reset();           // Reset for new level
private:
    int     NumZones;
    int     Row;
    char    *LandingZones;
};

#endif
```

```
#include "Gator.h"
bool Gator::DangerEatUs()
{
    bool    ret_val = false;

    if ((FrogPosition == 0) && bMouthOpen)    ret_val = true;

    return(ret_val);
}

void Gator::Move()
{
    FramesSinceAnimation++;

    if (FramesSinceAnimation >= (ANIMATION_SCALER * abs(FramesPerMove)))
    {
        FramesSinceAnimation = 0;
        bNeedsRepainting = true;
        if (bMouthOpen)
        {
            strcpy_s(Representation, sizeof(Representation), GATOR_PRESENTATION_C);
            bMouthOpen = false;
        }
        else
        {
            strcpy_s(Representation, sizeof(Representation), GATOR_PRESENTATION_O);
            bMouthOpen = true;
        }
    }
    Danger::Move();
}

void Snake::Move()
{
    Danger::Move();
}
```

```
#ifndef _GATOR
#define _GATOR
#include "Water.h"
#endif
```

```
#include "Globals.h"
#include "Danger.h"

void Danger::Move()
{
    FramesSinceLastMove++;

    if (FramesSinceLastMove >= abs(FramesPerMove))
    {
        FramesSinceLastMove = 0;
        Object::Move(SIGN(FramesPerMove));
    }
}

bool Danger::DetectTouching(int frogRow, int frogColumn)
{
    bool    ret_val = false;

    if (frogRow == Row)
    {
        // If the Danger hasn't wrapped
        if ((SCREEN_WIDTH - Width) > Column)
        {
            // Check for frog in bounding box of Truck. Must subtract one
            // because a length of 3 would occupy 17,18,19, and 19-17=2
            if (frogColumn >= Column && frogColumn <= (Column + Width - 1))
                ret_val = true;
        }
        else
            // The Danger is wrapping
            {
                int wrapAmount;

                wrapAmount = (Width+(int)Column-SCREEN_WIDTH);
                // Check the tail end
                if ((frogColumn >= Column) || (frogColumn < wrapAmount))
                    ret_val = true;
            }
    }

    return(ret_val);
}

bool Danger::DangerNeedsPainting()
{
    return (bNeedsRepainting);
}

void Danger::SetNeedsRepainting()
{
    bNeedsRepainting = true;
}
```

```

#ifndef _DANGER
#define _DANGER
#include "CConsole.h"
#include "Object.h"

#define VEHICLE_DANGERS (tTruck | tCar | tSemi)
#define RIDING_DANGERS (tBigLog | tShortLog | tGator | tSnake)

class Danger : public Object
{
protected:
    int      FrogPosition;           // If we carry a frog, where is it on us?
    int      FramesPerMove;          // The # times move must be called for
    advancement
    int      FramesSinceLastMove;    // Counter from 0 to framesPerMove
public:
    void      SetNeedsRepainting();
    virtual void Move();             // Moves the Danger x units along default
    direction. Also animates the character
    int      GetFrogPosition() {return FrogPosition;};
    void      SetFrogPosition(int x) {FrogPosition = x;};
    int      GetFramesPerMove() {return FramesPerMove;};
    bool      DangerNeedsPainting(); // True if Danger has moved
    int      GetVelocity();          // Returns Danger velocity
    int      GetSize();              // returns the length of the Danger
    virtual bool DangerEatUs(void) {return false;};
    bool      DetectTouching(int FrogRow, int FrogColumn); // Detect
    intersection of this Danger and the frog
    Danger();
    Danger(int objRow,
           int objColumn,
           int objWidth,
           Color fColor,
           Color bColor,
           int fpm):
        Object(objRow, objColumn, objWidth, fColor, bColor)
    { FramesPerMove = fpm; Danger::FramesSinceLastMove = 0; };
};
#endif

```

```

#ifndef _FLOATS
#define _FLOATS
#pragma once
#define BIGLOG_PRESENTATION "====="
#define GATOR_PRESENTATION_O ">=="
#define GATOR_PRESENTATION_C "+=="
#define SHORTLOG_PRESENTATION "LL"
#define SNAKE_PRESENTATION ">~"
#define ANIMATION_SCALER 6
#include "danger.h"

class Floats : public Danger
{
protected:
    bool    bCarringPlayer;

public:
    Floats()    {};
    Floats(int objRow, Color fColor, Color bColor, int framesPerMove, int
initOffset, char *objPresentation)
        :Danger(objRow, initOffset, (int)strlen(objPresentation), fColor, bColor,
framesPerMove)
        { strcpy_s(Representation, sizeof(Representation), objPresentation);
          bCarringPlayer = false; }
};

class ShortLog : public Floats
{
public:
    ShortLog(int objRow, Color fColor, Color bColor, int framesPerMove, int
initOffset)
        :Floats(objRow, fColor, bColor, framesPerMove, initOffset,
SHORTLOG_PRESENTATION)
        { Type = tShortLog; }
};

class FallingLog : public Floats
{
public:
    FallingLog(int objRow, Color fColor, Color bColor, int framesPerMove, int
initOffset)
        :Floats(objRow, fColor, bColor, framesPerMove, initOffset,
BIGLOG_PRESENTATION)
        { Type = tBigLog; }
};

class Gator : public Floats
{
public:
    Gator(int objRow, Color fColor, Color bColor, int framesPerMove, int
initOffset)
        :Floats(objRow, fColor, bColor, framesPerMove, initOffset,
GATOR_PRESENTATION_C)
        { Type = tGator;
          bMouthOpen = false;
          FramesSinceAnimation = 0;
        }
    bool    DangerEatUs();
    int     FramesSinceAnimation;
    bool    bMouthOpen;
    virtual void Move();
};

class Snake : public Gator
{

```



```
public:
    Snake(int objRow, Color fColor, Color bColor, int framesPerMove, int
initOffset)
    :Gator(objRow, fColor, bColor, framesPerMove, initOffset)
    { Width = (int)strlen(Representation); Type = tSnake; bMouthOpen = true;
      strcpy_s(Representation, sizeof(Representation), SNAKE_PRESENTATION);}
    void Move();
};
#endif
```

```

#ifndef _VEHICLE
#define _VEHICLE
#include "Object.h"

#define TRUCK_PRESENTATION "TTT"
#define CAR_PRESENTATION "CC"
#define SEMI_PRESENTATION "SSSS"

class RoadDanger : public Danger
{
protected:
    int Count; // Number of dangers per row. Will not end up implementing.
    int Spacing; // Number of spaces between each danger. Will not end up implementing.
public:
    RoadDanger() {}
    RoadDanger(int dRow, int dColumn, int dWidth, Color foreColor, Color backColor, int framesPerMove) : Danger(dRow, dColumn, dWidth, foreColor, backColor, framesPerMove) {}
};

class Vehicle : public RoadDanger
{
public:
    Vehicle(int objRow, int objCol, int objWidth, Color fColor, Color bColor, int velocity, int initOffset, char *textPresentation) : RoadDanger(objRow, initOffset, (int)strlen(textPresentation), fColor, bColor, velocity) {strcpy_s(Representation, sizeof(Representation), textPresentation);};
};

class Truck : public Vehicle
{
public:
    Truck(int objRow, Color fColor, Color bColor, int objVelocity, int initOffset) : Vehicle(objRow, initOffset, (int)strlen(TRUCK_PRESENTATION), fColor, bColor, objVelocity, initOffset, TRUCK_PRESENTATION) {Type = tTruck;};
};

class Car : public Vehicle
{
public:
    Car(int objRow, Color fColor, Color bColor, int framesPerMove, int initOffset) : Vehicle(objRow, initOffset, (int)strlen(CAR_PRESENTATION), fColor, bColor, framesPerMove, initOffset, CAR_PRESENTATION) {Type = tCar;};
};

class Semi : public Vehicle
{
public:
    Semi(int objRow, Color fColor, Color bColor, int framesPerMove, int initOffset) : Vehicle(objRow, initOffset, (int)strlen(SEMI_PRESENTATION), fColor, bColor, framesPerMove, initOffset, SEMI_PRESENTATION) {Type = tSemi;};
};

#endif

```

Senior Portfolio

Dave Coons

COIN 325: Java II

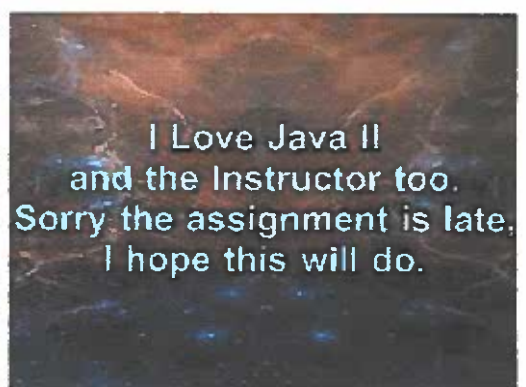
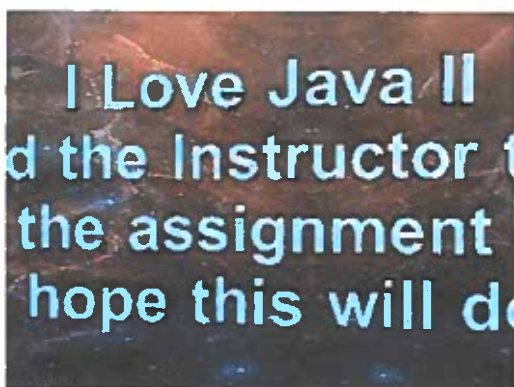
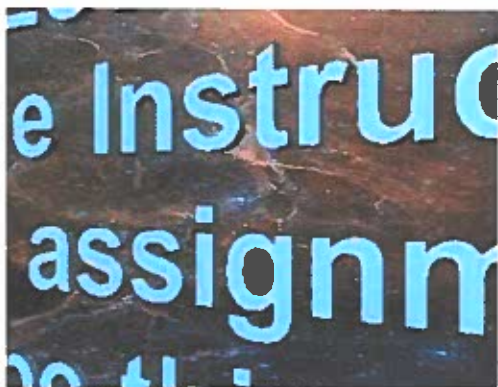
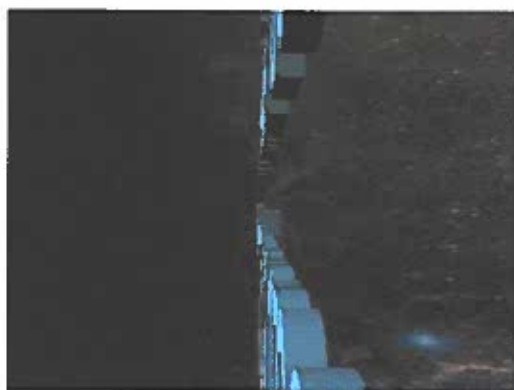
Flip Book

Grade Received: A

This is a simple Java applet that plays a 3D animation. The animation consists of flying by 3D text 'in space' with a 'space background' behind the letters. The text is actually a humorous attempt (or an attempt at being humorous) to appeal for a reduced late penalty for the assignment. The images were rendered in Autodesk's 3D Studio Max 8, then exported as a series of frames that were looped in Java to effect the animation. In order to keep the overall file size down (the assignment was to be uploaded upon completion), heavy compression was applied to each frame. Thus, the image quality may appear poor on the printout.

The CD also contains an installation script, as this was the 'final project' of the class.

NOTE: This class is substituted for Java I as I had previously taken Java at Trident Technical College while earning my Associate's in Applied Science: Computer Programming degree.



```

/*****
 *
 * Student Name:          Dave Coons
 *
 * Program Name:          assignment09
 *
 * Creation Date:         November 2009
 *
 * Last Modified Date:    November 2009
 *
 * COIN Course:           COIN 326: Java II
 *
 * Grade Received:        A
 *
 * Comments regarding design: Fairly straight forward 'flip-book' animation
 *
 *****/

```

```
package assignment09_test3;
```

```
import javax.swing.JFrame;
```

```
public class LogoAnimator
```

```

{
    // execute animation in a JFrame
    public static void main( String args[] )
    {
        LogoAnimatorJPanel animation = new LogoAnimatorJPanel();

        JFrame window = new JFrame( "Animator test" ); // set up window
        window.setDefaultCloseOperation( JFrame.EXIT_ON_CLOSE );
        window.add( animation ); // add panel to frame

        window.pack(); // make window just large enough for its GUI
        window.setVisible( true ); // display window

        animation.startAnimation(); // begin animation
    } // end main
} // end class LogoAnimator

```

```

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 *****/

```

```

/*
 * To change this template, choose Tools | Templates
 * and open the template in the editor.
 */

package assignment09_test3;

import java.awt.Dimension;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.awt.Graphics;
import java.net.MalformedURLException;
import javax.swing.ImageIcon;
import javax.swing.JPanel;
import javax.swing.Timer;
import java.net.URL;
import java.io.File;

public class LogoAnimatorJPanel extends JPanel
{
    private final static String IMAGE_NAME = "deitel"; // base image name
    protected ImageIcon images[]; // array of images
    private final int TOTAL_IMAGES = 121; // number of images
    private int currentImage = 0; // current image index
    private final int FRAMES_PER_SECOND = 30;
    private final int ANIMATION_DELAY = 1000 / FRAMES_PER_SECOND; // millisecond delay
    private int width; // image width
    private int height; // image height

    private Timer animationTimer; // Timer drives animation

    // constructor initializes LogoAnimatorJPanel by loading images
    public LogoAnimatorJPanel()
    {
        images = new ImageIcon[ TOTAL_IMAGES ];

        // load the images
        for ( int count = 0; count < images.length; count++ )
        {
            URL url = null;
            String filename = String.format("images\\Animation%04d.jpg", count);
            File file = new File(filename);
            try
            {
                url = file.toURL();
            }
            catch (java.net.MalformedURLException e)
            {
                System.err.println("Bad path to: "+filename);
            }

            System.out.println("Loading image : " + url);
            images[ count ] = new ImageIcon( url );
        }

        // this example assumes all images have the same width and height
        width = images[ 0 ].getIconWidth(); // get icon width
        height = images[ 0 ].getIconHeight(); // get icon height
    } // end LogoAnimatorJPanel constructor

    // display current image
    public void paintComponent( Graphics g )
    {
        super.paintComponent( g ); // call superclass paintComponent
    }

```

```
images[ currentImage ].paintIcon( this, g, 0, 0 );

// set next image to be drawn only if Timer is running
if ( animationTimer.isRunning() )
{
    // Minus 2 because minus 1 is last image, and we will increment past
    // current image after if statement, So (TOTAL_IMAGES -1) - 1
    if (currentImage == (TOTAL_IMAGES - 2))
        animationTimer.setDelay(ANIMATION_DELAY * FRAMES_PER_SECOND * 3);
    else
        animationTimer.setDelay(ANIMATION_DELAY);
    currentImage = ( currentImage + 1 ) % TOTAL_IMAGES;
}
} // end method paintComponent

// start animation, or restart if window is redisplayed
public void startAnimation()
{
    if ( animationTimer == null )
    {
        currentImage = 0; // display first image

        // create timer
        animationTimer = new Timer( ANIMATION_DELAY, new TimerHandler() );

        animationTimer.start(); // start Timer
    } // end if
    else // animationTimer already exists, restart animation
    {
        if ( ! animationTimer.isRunning() )
            animationTimer.restart();
    } // end else
} // end method startAnimation

// stop animation Timer
public void stopAnimation()
{
    animationTimer.stop();
} // end method stopAnimation

// return minimum size of animation
public Dimension getMinimumSize()
{
    return getPreferredSize();
} // end method getMinimumSize

// return preferred size of animation
public Dimension getPreferredSize()
{
    return new Dimension( width, height );
} // end method getPreferredSize

// inner class to handle action events from Timer
private class TimerHandler implements ActionListener
{
    // respond to Timer's event
    public void actionPerformed( ActionEvent actionEvent )
    {
        repaint(); // repaint animator
    } // end method actionPerformed
} // end class TimerHandler
} // end class LogoAnimatorJPanel

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 *****/
```

```
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*****/
```




Senior Portfolio

Dave Coons

COIN 333: Applied Systems

Assembly Adder

Grade Received: A

This very simple program will prompt the user for two numbers and add them together, printing out the result. This assembly program is written in MIPS assembly, and is run on a MIPS simulator.



Console

Enter a number: 10
Enter a number: 20
30



```

#*****
# Student Name: Dave Coons *
# Program Name: Adder.s *
# Creation Date: March 2009 *
# Last Modified Date: March 2009 *
# COIN Course: COIN 333: Applied Systems *
# Grade Received: A *
# Comments regarding design: MIPS Assembly simple adder *
#*****/

```

```
# Adder
```

```

.data # Data declaration section
FIRST: .word 0
SECOND: .word 0
ANSWER: .word 0
ENTER: .asciiz "Enter a number: "
PRINT_INT: .word 1 # $a0 has integer
PRINT_STRING: .word 4 # $a0 has string
READ_INT: .word 5 # integer result in $v0
READ_STRING: .word 8 # $a0 buffer, $a1 length
.text

```

```
main: # Start of code section
```

```

#void main () {
#   int first, second, answer;//you do NOT need
#//to declare these as variables in MIPS, just use registers

```

```

#   cout <<"Enter a number: ";
lw $v0, PRINT_STRING
la $a0, ENTER
syscall

```

```

#   cin >>first;
lw $v0, READ_INT
syscall
sw $v0, FIRST

```

```

#   cout <<"Enter a number: ";
lw $v0, PRINT_STRING
la $a0, ENTER
syscall

```

```

#   cin >>first;
lw $v0, READ_INT
syscall
sw $v0, SECOND

```

```

#   answer=add2(first, second);
lw $a0, FIRST
lw $a1, SECOND
la $a2, ANSWER
jal add2

```

```

#   cout <<answer;
lw $a0, ANSWER
lw $v0, PRINT_INT
syscall

```

```

# Terminate execution
li $v0, 10
syscall

```

```

add2: #void add2Ptr (int x, int y, int *ans)
#           $a0   $a1   $a2

```

```
add $t0, $a0, $a1
sw  $t0, ($a2)
jr  $ra
```

```
# END OF PROGRAM
```

Coins 332 Ethics Paper

Grade Received: A

This ethics paper was a 'response' to an article castigating the RIAA for its actions against users of services such as the former Napster. The article expresses outrage towards the RIAA for their actions against college students, and sympathy for the defendants in those cases.



The Misnomer of File Sharing

Thursday, November 19th 2007

By: Dave Coons

Applied Networking

COIN 332-61

Instructor: Yu-Ju Lin

Copyright law is a very tenuous concept. It's obvious why stealing or murder is wrong, but copyright violations? It seems as though it hurts no one. However, there is a clear victim and clear damage is done, even though the actual victim may not even realize they are being harmed.

Copyright violations are so necessary to a functional economy that they are protected by the United States Constitution. The US Constitution grants the national congress the power

To promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries; (US Constitution, Article I, Section 8, Clause 8)

This concept was ingrained in the constitution because the Founding Fathers realized that without copyright protection, there is no incentive to create.

A popular defense of file 'sharing' is that it harms no one. The very notion of 'sharing' congers up friendly imagery. However, some take issue with the term 'sharing'. Lars Ulrich, drummer for the popular band *Metallica*, described sharing to a congressional inquiry as follows

If I share a sandwich, I have only half a sandwich. If I share my car, I loose the use of my car for a day. (paraphrasing)

What was really happening, said Ulrich, is file ***duplicating***. There in lies the rub. If you share a file with someone, you are not dividing a single resource among multiple parties (which is allowed under many licenses). You are multiplying a single resource for multiple parties. Every time a copyrighted work is duplicated, there is one less potential sale for the copyright owner. The danger to society is that, left unchecked, copyright duplication can reduce the revenue potential of creativity to the point to where it ceases.

The RIAA is therefore presented with a very difficult task. They have to limit file duplication as much as possible. In most illicit activities, the traditional role is to go after the provider. Governments usually target the drug dealers, not the drug users. In traditional copyright violation, an individual or company makes illegal copies of an item and sells them for profit. In this case, if you stop that individual, you remove all the potential lost sales that 'pirate' would have eliminated. File sharing networks pose a different problem. As in prostitution, the majority of the crimes are between individuals. Therefore, to stop the crime, you must target the small-time individuals.

The article gives several examples of the RIAA pursuing end user copyright violators. A college student, Sarah Barg at the University of Nebraska-Lincoln, was sued by the RIAA for \$3000 for the possession (and presumed sharing) of 381 copyrighted tracks from the 80's and 90's. This article portrays this as an outrage, saying that these poor college students would have to get second jobs to pay these 'hefty fee(s)'. Another individual named Johnson downloaded (and presumably shared) 100 songs and had to pay the fine from his college fund. Poor Johnson will have to work extra summer jobs to pay the fine.

Boo hoo.

Sarah Barg paid \$3000 for 381 songs. That's about \$7.87 a song, a little steep, but not outrageous. And Sarah Barg, being a college student, presumably knew that file swapping of copyrighted material is not legal. Johnson had to get extra summer jobs to pay for the music he 'stole'. Poor guy, having to pay for what he received.

The article then descends into moral disarray. The copyright violations are compared to speeding marginally over the speed limit. This is a very good comparison, since both practices are illegal for esoteric reasons. However, the author then goes on to say that the fines are a worse fate, since "one is far more likely to beat the speeding fine than an RIAA lawsuit in court." What ever happened to taking responsibility for ones actions? If you knowingly break the law, take your punishment like a man!

The basic premise for file sharing is that materials possessions should be protected by law, but information is free. This sounds good, until you think it through to its logical conclusion. If movies and songs and books are all freely shared information, what's the incentive to create them? George Lucas, creator of the Star Wars and Indiana Jones franchises, commented that the age of big budget films will soon draw to a close. Why? Because copyright infringement will soon make such monumental investments cost prohibitive. Why would a film producer spend \$200 million to make a movie if everyone is going to steal it for free?

It is not a coincidence that file sharing is very prevalent on college campuses. In modern colleges, communism and socialism reign supreme. College students are old enough to get things from society, but not old enough to have enough money for society to take. They think they have a right to free music because the big music industry charges 'too

much'. However, their Robin Hood vigilantism of taking from the music industry and giving to the poor will only serve to limit the quality and quantity of the music they steal. One can only hope that some of these college students will develop into musicians (or programmers or writers or what have you) and have others steal their works. Their view of Robin Hood might just change.